

Abstract:

This project analyzes large scale shot data from the NBA and NHL to develop comparable performance metrics across two different sports. The NBA dataset comes from the official NBA Stats API and the ESPN API, representing the industry standard for basketball analytics. NHL data is sourced from MoneyPuck, a transparent repository of shot information with expert validated variable definitions and corrections for coordinate inaccuracies. Using these datasets, we aim to construct expected field goal (xFG) and expected goal (xG) models with shot heat maps, analyze shot difficulty, and evaluate player performance through Points Over Expected (POE). Our goals include building predictive models for shot success, identifying overrated and underrated players through residual analysis, clustering players into shot archetypes, and developing a cross-sport normalization framework that allows meaningful comparison between NBA and NHL scoring. Ultimately, this project seeks to produce a combined analytical report for evaluating offensive performance across two major professional sports.

Introduction:

Shot analysis modeling has become central to modern sports analytics, yet most research focuses on a single sport. Our project explores whether expected scoring frameworks and shot analysis can be compared across the NBA and NHL by building parallel models using high quality shot data. We are interested in this topic because we are all into sports but analyzing two different sports with two different shot expectations as far as pace, volume, and expected shot conversion rate presents a fun challenge while being different than just your average sports analysis. Understanding how shot difficulty translates across sports may reveal deeper insights into player value, player archetypes and decision making.

Methods:

NBA data has been collected using the NBA Stats API and ESPN API, which provides, among other things, shot coordinates, shooter identity, defender proximity, shot type, and other metadata. NHL data has been obtained from MoneyPuck's public shot dataset, which includes coordinates, angles, rebounds, goalie interaction variables, and many other factors that involve each shot. We plan on using these data sets to create cross-over metrics and analysis to provide better insight into either sport respectively. We will build a logistic regression model to estimate expected field goal probability (xFG) in the NBA and an expected goal probability (xG) for the NHL. We may use other regression models to better understand and improve these metrics. We plan to provide residual analysis to identify players who over/underperform expectations, a Shot Difficulty Index (SDI) quantifying shot complexity, and (3) role-aware GMM clustering to identify player shot archetypes. We also expect to create Points Over Expected (POE) metrics for both NBA and NHL shots to better understand and analyze player comparison and performance across leagues. We will be working in both Python and R using any and all packages available.

There are a few challenges we expect to face when taking on this project. Key challenges include normalizing metrics across sports with different scoring scales, handling large datasets efficiently, and aligning salary-based POE metrics across leagues with different cap structures.

Expected Outcomes:

If modeling proceeds as planned, we expect to produce accurate xFG and xG models that identify the key predictors of shot success in each sport. We anticipate discovering players who consistently outperform expected scoring, revealing insights into finishing skill and shot selection. The Shot Difficulty Index should highlight structural differences between NBA and NHL shot environments, while clustering will uncover distinct shooter archetypes. Finally, we expect to develop a normalized POE metric that enables meaningful cross-sport comparison of scoring efficiency, offering an insightful perspective on player value across two major professional leagues.

Group Considerations: (Include what analysis you plan to do and if there is cross-over please make the appropriate percentages) (Also if you want help with something in this project throw my name next to yours and I'll help - Kanaan)

Data Collection (Kaleb 50% Drianna 50%)

Final poster Presentation (Kanaan 33% Kaleb 33% Drianna 33%)

NBA and NHL cross-over modeling and analysis (Kanaan 33% Kaleb 33% Drianna 33%)

NHL POE and xG prediction (Kanaan)

NHL prediction modeling (Kanaan)